

QIAN GORLOS, China

WMO: 509490

Lat: 45.0908N

Lon: 124.8670E

Elev: 136

StdP: 99.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-24.3	-31.6	0.2	-25.2	-29.4	0.3	-22.1	6.9	-9.3	6.2	-9.2	1.3	220	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	31.9	21.5	30.5	21.5	29.2	21.1	24.8	28.7	23.9	27.6	23.0	26.8	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.7	27.4	22.7	17.7	26.5	21.7	16.7	25.6	76.5	28.8	72.9	27.6	69.3	26.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.5	5.6	-28.2	34.5	4.3	1.7	-31.3	35.7	-33.8	36.7	-36.2	37.7	-39.4	38.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-28.4	25.8	4.2	1.2	-31.5	26.7	-34.0	27.4	-36.3	28.1	-39.4	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-15.6	-10.3	-1.6	8.4	16.1	22.1	24.2	22.5	16.5	7.7	-3.7	-13.1
	DBStd	14.67	4.74	5.65	5.63	4.89	4.09	3.28	2.29	2.54	3.64	4.84	6.16	4.97
	HDD10.0	3043	793	569	360	87	3	0	0	2	103	411	716	
	HDD18.3	4916	1051	803	617	299	94	6	1	3	76	331	661	974
	CDD10.0	1652	0	0	1	39	193	362	441	388	197	31	1	0
	CDD18.3	482	0	0	0	2	26	118	183	132	22	1	0	0
	CDH23.3	3806	0	0	0	20	260	989	1455	911	164	6	0	0
	CDH26.7	1099	0	0	0	2	71	343	436	224	22	1	0	0
Wind	WSAvg	2.5	2.1	2.4	3.0	3.4	3.1	2.5	2.2	2.1	2.2	2.5	2.6	2.2
Precipitation	PrecAvg	438	1	3	6	17	33	76	143	91	44	17	4	2
	PrecMax	593	6	10	22	151	84	161	252	200	120	62	16	10
	PrecMin	243	0	0	0	0	6	14	50	23	9	0	0	0
	PrecStd	76	1	3	6	27	22	34	49	51	28	15	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	6.6	17.3	26.0	31.2	34.6	33.4	31.7	29.0	24.0	13.8	4.1
		MCWB	-3.6	0.6	7.8	12.8	17.0	20.0	23.9	23.6	19.8	14.5	8.4	1.6
	2%	DB	-3.9	3.7	12.7	22.4	28.7	32.2	31.7	30.3	27.0	20.4	10.8	-0.8
		MCWB	-6.2	-0.7	4.9	10.9	15.8	19.5	23.2	22.8	18.6	13.0	6.3	-3.0
	5%	DB	-6.2	1.2	9.9	20.0	26.3	30.4	30.4	29.0	25.3	17.7	7.9	-3.3
		MCWB	-8.1	-2.5	3.4	9.8	14.8	19.5	23.0	21.8	17.5	11.3	4.3	-5.4
	10%	DB	-8.3	-1.4	7.3	17.5	23.9	28.5	29.1	27.8	23.5	15.5	4.8	-5.8
		MCWB	-10.0	-4.5	1.6	8.9	13.8	19.1	22.5	21.4	16.3	10.1	1.6	-7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.5	1.6	8.3	14.5	19.3	23.0	26.6	25.6	22.2	16.2	9.4	1.5
		MCDB	-1.0	5.4	16.1	22.4	27.8	29.3	30.9	29.9	26.7	20.3	12.7	3.6
	2%	WB	-6.1	-0.5	5.6	12.2	17.4	22.0	25.2	24.6	20.3	14.0	6.6	-2.7
		MCDB	-4.0	3.4	12.0	20.8	25.3	28.6	29.2	28.5	24.4	19.6	10.2	-1.0
	5%	WB	-8.1	-2.6	3.7	10.6	16.0	21.2	24.4	23.7	18.9	12.5	4.2	-5.4
		MCDB	-6.3	1.2	9.4	18.9	23.7	27.0	28.2	27.1	23.3	16.8	7.5	-3.4
	10%	WB	-10.0	-4.4	1.8	9.1	14.8	20.3	23.6	22.7	17.6	10.6	1.7	-7.6
		MCDB	-8.4	-1.5	6.8	16.4	22.6	25.9	27.4	26.1	22.0	14.9	4.6	-5.9
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.9	10.3	11.3	10.6	9.3	8.0	8.6	11.1	10.4	8.7	9.0
		MCDBR	11.7	12.4	13.8	15.0	14.0	12.3	9.9	10.2	13.1	13.2	11.4	10.7
		MCWBR	9.9	9.3	8.2	7.7	5.8	4.5	3.9	4.3	5.8	7.2	7.9	9.0
	5% WB	MCDBR	11.6	12.0	13.1	13.2	11.7	9.6	7.9	8.1	10.4	11.7	10.9	10.6
		MCWBR	9.9	9.1	8.2	7.6	5.7	4.1	3.9	4.3	5.7	7.4	8.1	9.1
Clear-Sky Solar Irradiance	taub	0.315	0.335	0.411	0.509	0.582	0.643	0.595	0.525	0.465	0.450	0.368	0.326	
	taud	2.125	2.109	1.963	1.788	1.687	1.656	1.796	1.949	2.006	1.937	2.067	2.093	
	Ebn at Noon	764	825	802	753	712	670	699	734	741	676	681	695	
	Edh at Noon	95	118	158	205	235	243	209	173	150	140	101	89	
All-Sky Solar Radiation	RadAvg	2.13	3.22	4.34	5.18	5.76	5.66	5.33	5.01	4.45	3.14	2.09	1.73	
	RadStd	0.20	0.23	0.24	0.38	0.42	0.47	0.49	0.33	0.36	0.19	0.16	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page